Continuity and Differentiability Test

Test the differentiability of the following functions:

1. $f(x) = \begin{cases} Cosx & , x > 0 \\ -Cosx & , x < 0 \end{cases}$ at x = 0
2. $f(x) = \begin{cases} x^2 Sin(1/x) & , x \neq 0 \\ 0 & x = 0 \end{cases}$ at x = 0
3. $f(x) = |x|$ at $x = 0$
4. $f(x) = \begin{cases} x^2 - 16x & , x < 9 \\ 12\sqrt{x} & , x \geq 9 \end{cases}$

   Is $f(x)$ continuous at $9=x$? Determine whether $f(x)$ is differentiable at x = 9.
5. $f(x) = \begin{cases} x^2 & , x \leq 1 \\ \sqrt{x} & , x > 1 \end{cases}$

   Is $f(x)$ continuous at x = 1? Determine whether $f(x)$ is differentiable at x = 1.
6. $f(x) = \begin{cases} xCos(1/x) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ at x = 0
7. Show that $f(x) = \begin{cases} x^2 + 1 & , x \leq 1 \\ x & , x > 1 \end{cases}$ is not continuous and differentiable at x = 1.